

PORTFOLIO RISK REVIEW

Credit Risk Analysis

Identifying high-risk borrower characteristics and evaluating lending strategies to improve portfolio quality

Executive Summary · 255,347 loans · \$32.58B portfolio

Where the Portfolio Stands Today

255,347

Total Loans

Full portfolio, 2024

11.61%

Default Rate

~1 in 9 borrowers

\$32.58B

Total Exposure

Full lending volume

\$4.29B

Expected Loss

13.2% of exposure

Key Findings

- Risk is multidimensional: no single variable strongly predicts default — Age and Interest Rate are the strongest signals, but each explains only a small share of outcomes.
- Income is the clearest lever: low-income borrowers default at 17.4%, nearly double the 9.0% rate of the highest-income segment — while loan sizes are nearly identical across all income levels (~\$127.6K).
- A combined loan-sizing and rejection strategy can reduce expected credit losses by roughly \$346M (8% of current expected loss) while touching under 5% of total lending volume (\$1.51B of \$32.58B).

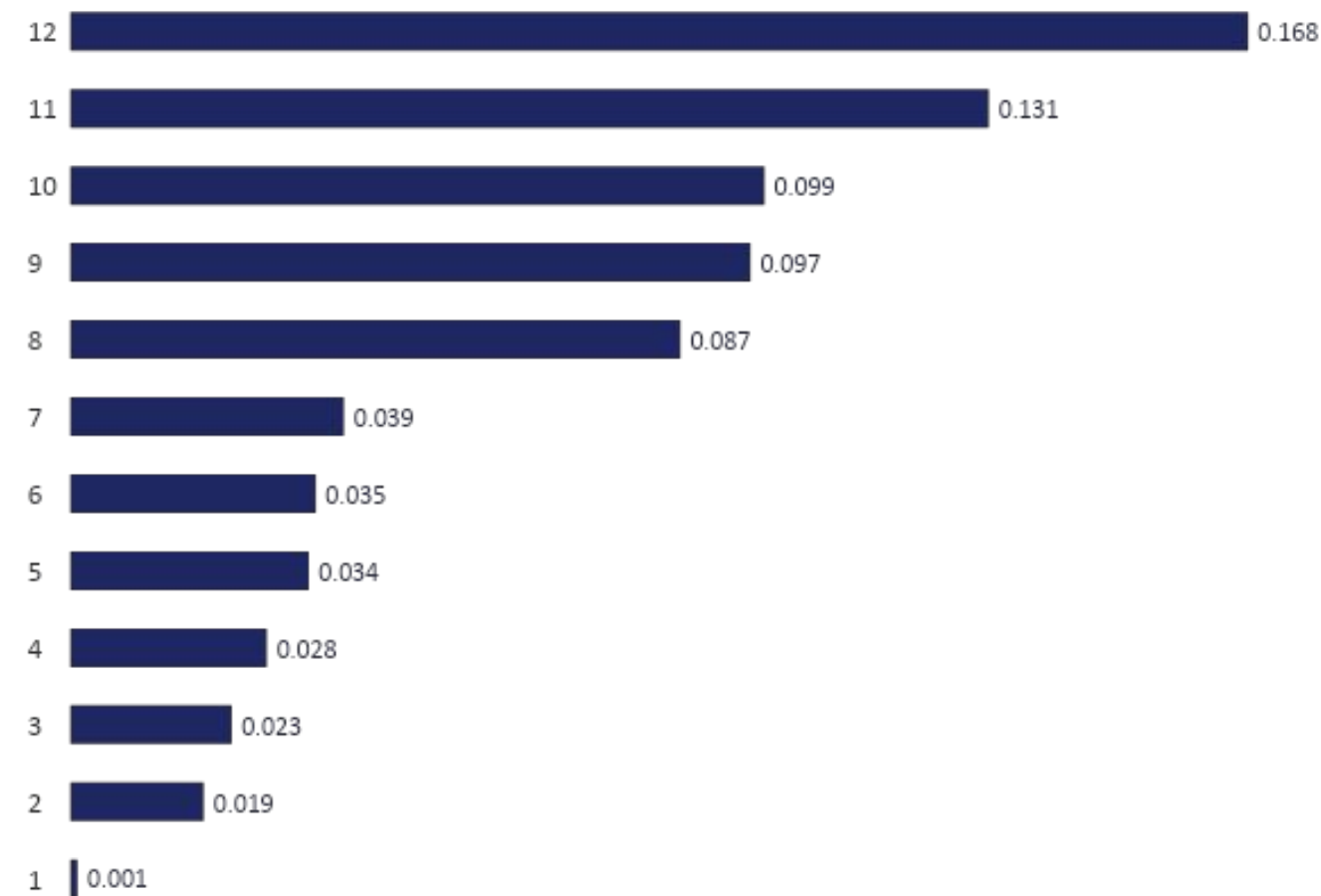
RECOMMENDATION

Hybrid Lending Strategy

- Cap loans 10% for all low-income borrowers
- Cap loans 20% for the ultra-high-risk sub-segment (low income + credit score < 350 + DTI > 0.60)
- Reject the narrow segment meeting all of: credit score < 400, bottom-25% income, DTI > 55%

~\$1.51B volume affected · ~\$346M expected loss reduction

What Drives Default Risk?



What the Data Tells Us

Age & Interest Rate are the strongest signals — older borrowers and lower rates correlate with lower default risk.

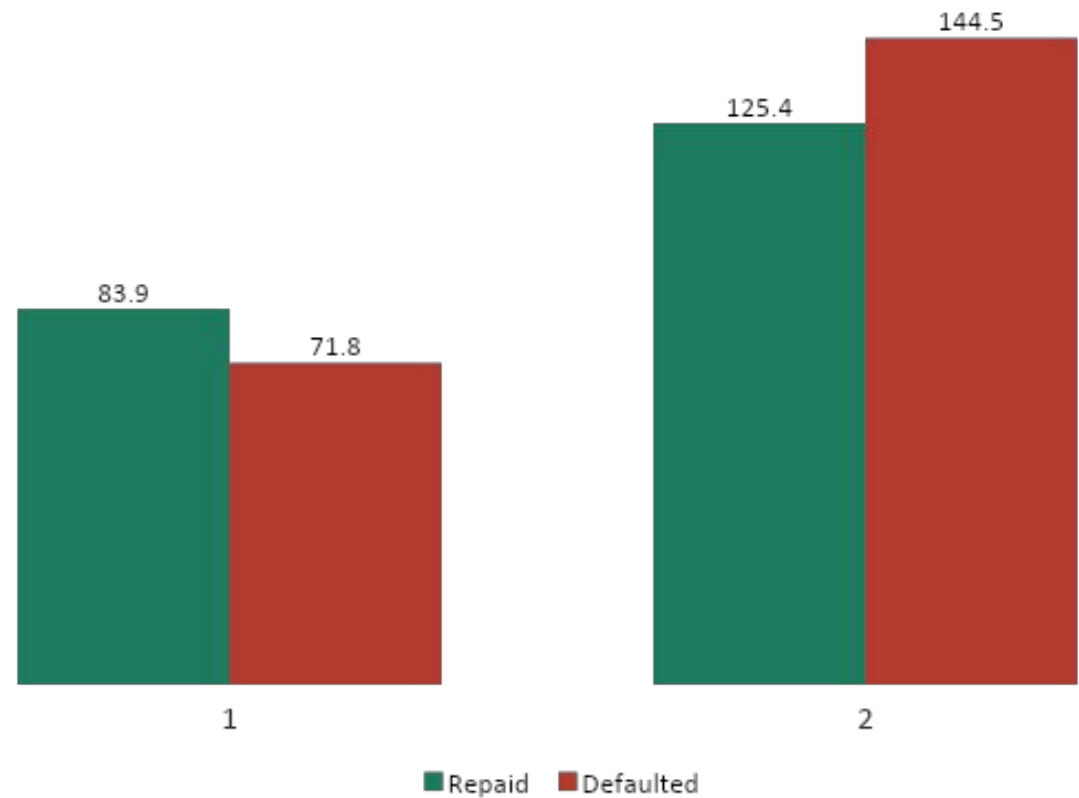
Income & tenure show a moderate protective effect; longer-employed, higher-income borrowers default less.

Credit Score has a weak standalone relationship (0.034) — it ranks 8th of 12 factors and is not a dominant predictor in this portfolio.

No single variable strongly predicts default. Risk is multidimensional — it takes a combination of factors to identify true high-risk borrowers.

Who Are Our Defaulting Borrowers?

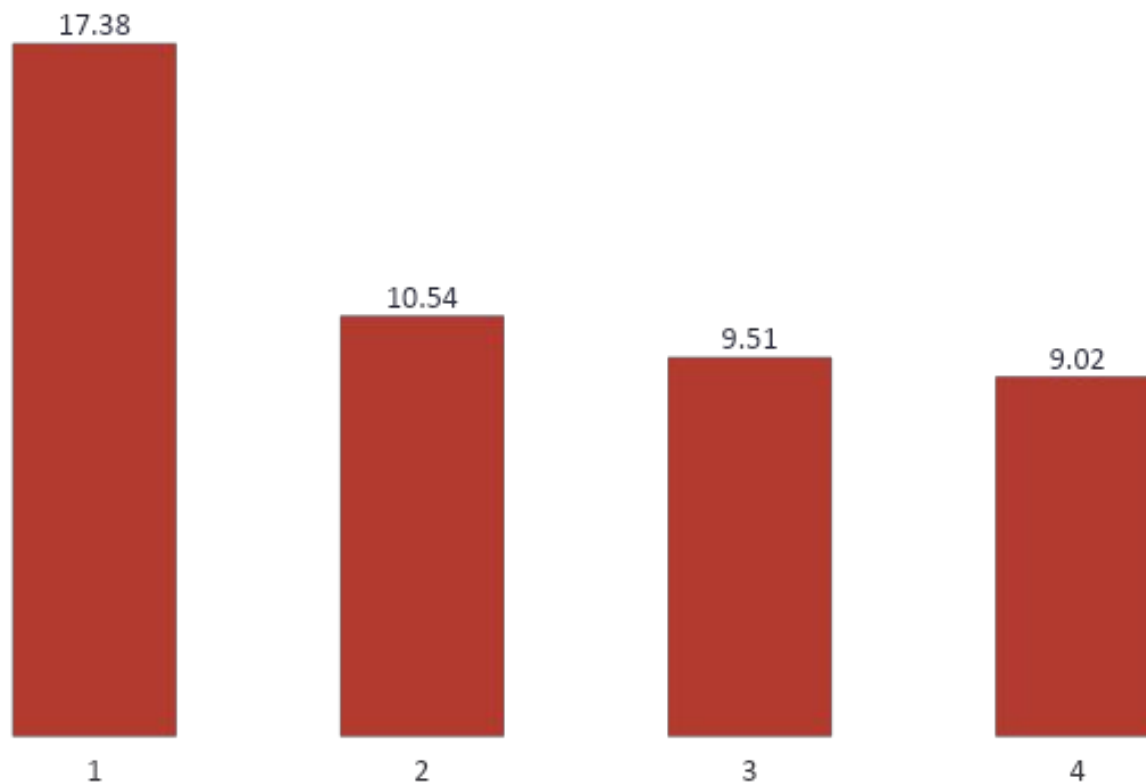
Defaulted borrowers share a consistent financial mismatch: lower income, larger loans, higher repayment pressure.



Key Differences

- Income gap:** Defaulters earn \$12.1K less on average (\$71.8K vs \$83.9K).
- Loan size gap:** Defaulters borrow \$19.2K more on average (\$144.5K vs \$125.4K).
- Credit score:** Slightly lower (559 vs 576) — a 17-point gap that alone is not a strong differentiator.
- Employment:** Unemployed borrowers default at 13.6% vs 9.5% for full-time employees.

Which Borrowers Are Highest Risk?



Inverse Relationship: Income vs. Default

- Low-income borrowers default at 17.4% — nearly 2x the 9.0% rate of the highest-income segment.
- Average loan size is nearly identical across all four income segments (~\$127.6K) — loan amounts are not risk-adjusted by income.
- Average credit score is also flat across income segments (~574) — income, not credit score, separates risk here.

Low-income segment: 63,837 borrowers · \$8.15B exposure · 17.38% observed default rate.

How Well Does Our Model Perform?

Evaluated on a 51,070-loan holdout test set (\$6.53B volume, ~20% of the portfolio) at an approval threshold of 0.20

	Predicted: Decline	Predicted: Approve
Actual: Defaulter	2,624 correctly caught	3,307 missed (risk approved)
Actual: Non-Defaulter	5,905 wrongly declined	39,234 correctly approved

Model correctly flags 44.2% of eventual defaulters (2,624 of 5,931) while approving 83% of all applications — a deliberately volume-friendly setting rather than an aggressive risk filter.

0.751

ROC-AUC

88.6%

Accuracy

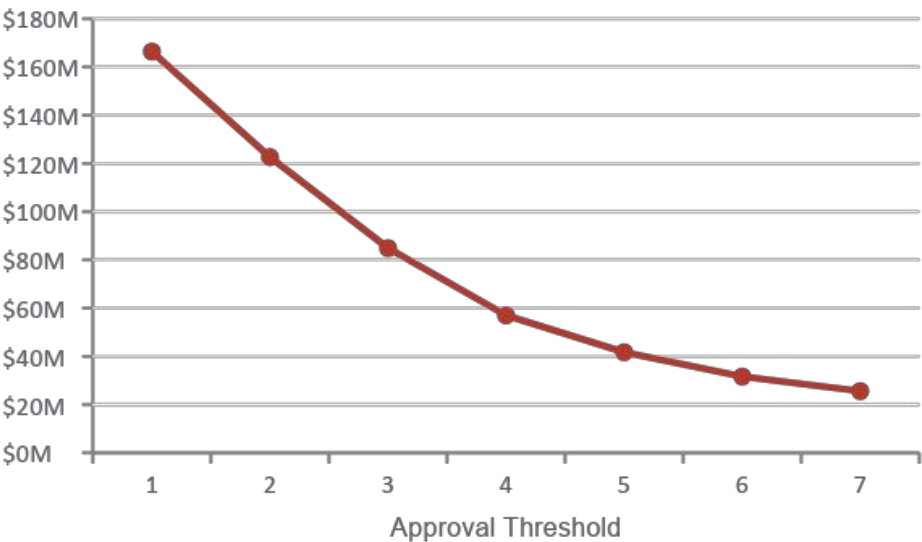
83%

Approval Rate

7.8%

Risk in Approved Portfolio (from 11.6%)

Simulated Profit by Approval Threshold

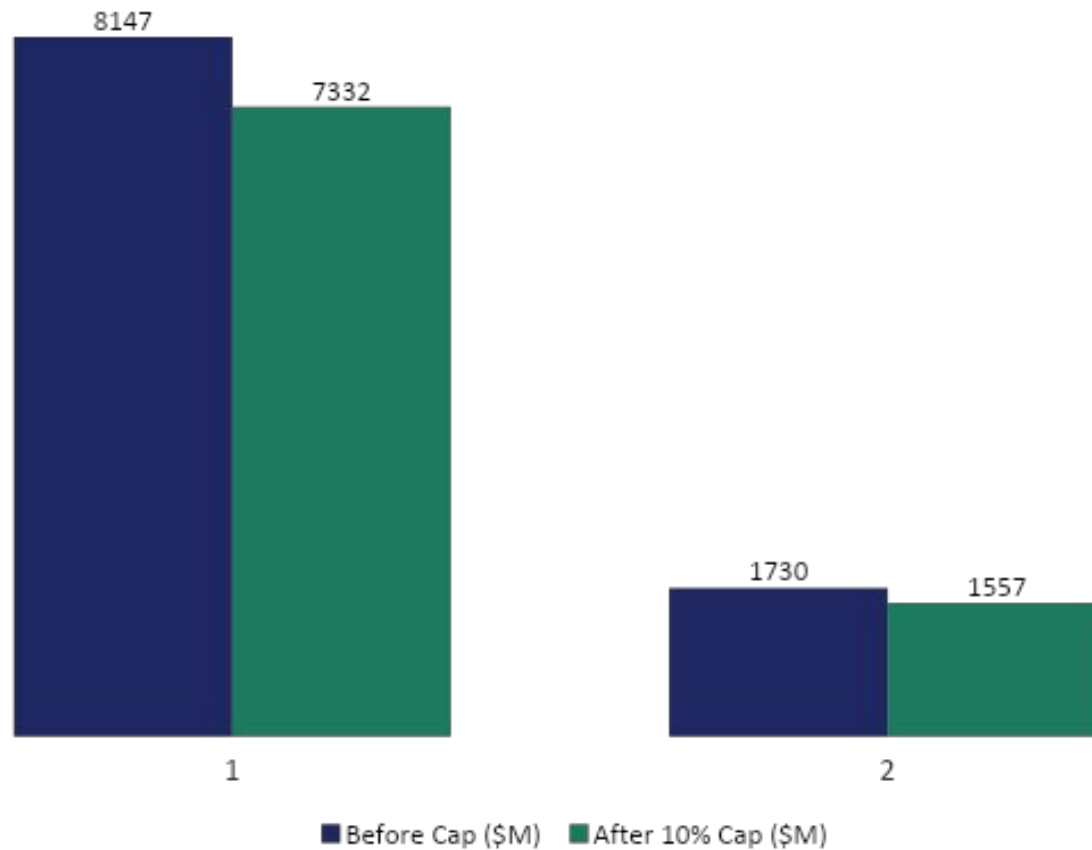


Lower thresholds approve more volume and raise profit but let more risk through; \$166.4M at 0.20 is the highest simulated profit tested.

1

STRATEGY 1

Loan Cap on Low-Income Borrowers



Impact of the Cap

Exposure reduction: \$814.7M

Expected loss reduction: \$173.2M

Efficiency: 21.3 cents saved per \$1 of lending removed

Avg loan per borrower: \$127,624 → \$114,862

Borrowers affected: 63,837 (all low-income segment)

Loan amounts are uniform across income groups, so capping low-income borrowers directly reduces concentrated risk without restructuring the portfolio.

Targeted High-Risk Segment Cap

Ultra-targeted filter: **Low Income + Credit Score < 350 + DTI > 0.60** → 2,109 borrowers · \$268.3M exposure · 20.6% observed default rate

RECOMMENDED

Scenario A — 10% Cap

Exposure reduced

\$26.8M

Expected loss reduced

\$6.7M

Efficiency

25.1%

Scenario B — 20% Cap

Exposure reduced

\$53.7M

Expected loss reduced

\$13.5M

Efficiency

25.1%

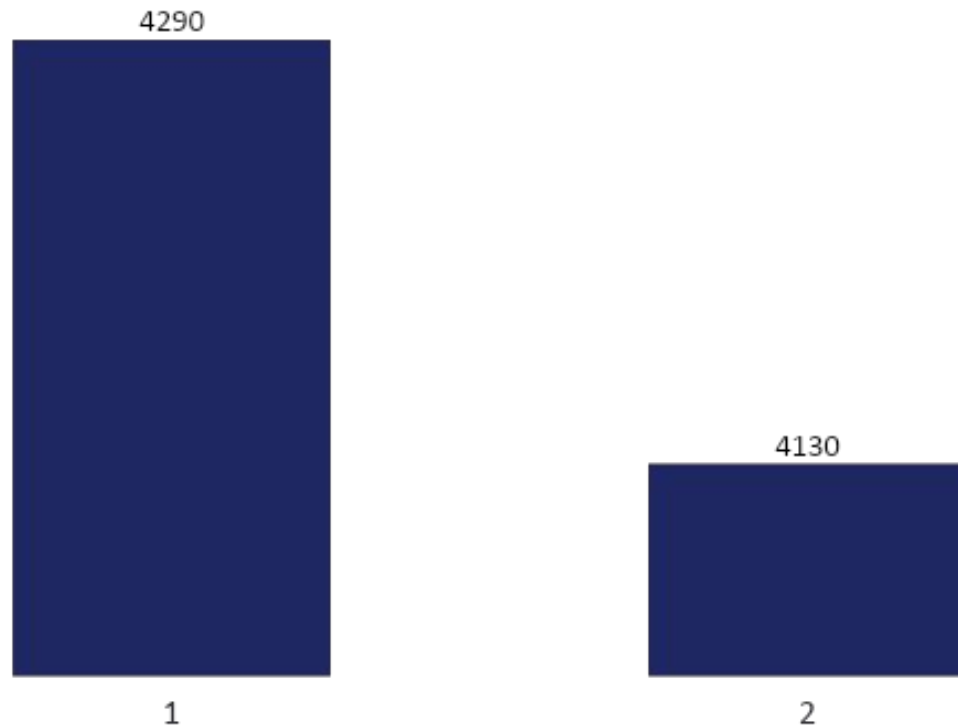
Why 20%, Not 10%

- Both scenarios recover the same 25.1 cents of expected loss per dollar of lending removed — efficiency doesn't fall as the cap deepens.
- The deeper 20% cap doubles the loss reduction (\$13.5M vs \$6.7M) for the same 2,109-borrower segment.
- This is the scenario carried into the final combined recommendation.

Targeting only the highest-risk borrowers produces a higher risk-reduction efficiency than the broad low-income cap alone (25.1% vs 21.3%) — though the total dollar impact is smaller because the segment itself is much smaller.

High-Risk Borrower Rejection Policy

Reject applications meeting all three: **Credit Score < 400** · **Bottom 25% Income** · **DTI > 55%**



Portfolio Impact

5,071 applications rejected — a small fraction of the portfolio

\$645.6M lending volume removed

\$159.7M expected loss avoided (3.7% of total)

11.61% → 11.44% portfolio default rate (-0.17 pp)

A narrow, precisely targeted policy: rejecting the highest-risk 2% of applications avoids a disproportionate share of expected losses.

THE RECOMMENDATION

A Combined, Consistent Hybrid Strategy

1

Apply a 10% loan cap to all low-income borrowers

Reduces \$814.7M in exposure and \$173.2M in expected loss across 63,837 borrowers.

2

Apply a 20% loan cap to the ultra-high-risk sub-segment

Credit Score < 350 + DTI > 0.60: an additional \$53.7M exposure and \$13.5M loss reduction across 2,109 borrowers.

3

Reject applications with Score < 400 + bottom-25% income + DTI > 55%

Avoids \$159.7M in expected losses by declining only 5,071 of the highest-risk applications.

Combined Impact (Strategies 1 + 2 + 3, applied consistently)

~\$1.51B

Lending Volume Affected

\$814.7M + \$53.7M + \$645.6M

~\$346.4M

Expected Loss Reduction

8.1% of current \$4.29B expected loss

~\$3.94B

Resulting Expected Loss

Down from \$4.29B baseline

11.44%

Portfolio Default Rate

Only Strategy 3 shifts default count directly

Note: Strategy 2's 2,109-borrower segment sits within Strategy 1's low-income population; the 20% cap is applied in place of (not in addition to) Strategy 1's 10% cap for that subset.